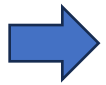


Module 14)

python – collection , function and modules



1.Accessing List

1. Understanding how to create and access elements in a list.

- List is a collection data type in python. We can store multiple values single variable it is called list.
- Denoted by []
- List is mutable.
- List is orderable.
- List allow duplicate values.

• Creating a List

- You can create a list by placing items (elements) inside square brackets [], separated by commas:
- `fruits = ["apple", "banana", "cherry"]`

• Accessing Elements in a List

1. Access by Index

Python uses zero-based indexing, which means the first element is at index 0 .

2. Access from the End (Negative Indexing)

You can also use negative numbers to access elements from the end of the list .

3. Slicing a List

You can get a sublist using slicing .

2. Indexing in lists (positive and negative indexing).

- A list in Python is an ordered collection of items. You can access individual items using indexes.

- **Positive Indexing**

Indexing starts at 0.

The first element has index 0, the second has index 1, and so on.

Example :

```
fruits = ["apple", "banana", "cherry", "date"]
```

```
print(fruits[0]) # Output: apple
```

```
print(fruits[1]) # Output: banana
```

```
print(fruits[2]) # Output: cherry
```

```
print(fruits[3]) # Output: date
```

- **Negative Indexing**

Negative indexes count from the end of the list.

-1 is the last item, -2 is the second last, etc..

Example

```
fruits = ["apple", "banana", "cherry", "date"]
```

```
print(fruits[-1]) # Output: date
```

```
print(fruits[-2]) # Output: cherry
```

```
print(fruits[-3]) # Output: banana
```

```
print(fruits[-4]) # Output: apple
```

3. Slicing a list: accessing a range of elements.

➤ Syntax

```
list[start:stop]
```

- start: index to begin the slice

- stop: index to end the slice

- Example

```
fruits = ["apple", "banana", "cherry", "date", "elderberry"]
```

```
print(fruits[1:4]) # Output: ['banana', 'cherry', 'date']
```

```
print(fruits[0:3]) # Output: ['apple', 'banana', 'cherry']
```

```
print(fruits[-3:-1]) # Output: ['cherry', 'date']
```